

Embedded

Using an RTOS (preferably Zephyr OS), create a firmware to print received CAN messages to a UART (CDC or physical). The output should include timestamp, CAN ID, DLC and data. The program should also:

- Be build time configurable to allow control of the CAN peripheral setup, task periods and CAN IDs used below.
- Three TX messages sent at build time defined periods. The data for these messages should be randomly generated.
- An optional RX message ID to start printing. If undefined the program should start immediately.
- An optional RX message ID to stop printing. If undefined the program should continue indefinitely unless halted.
- An optional RX message ID to print 'hello specialized'.

Include some unit tests to demonstrate this functionality.